EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 06

Basic Maths and Calculus

A.P., G.P. and Straight-Line Graphs

By — Excellence Programme

9 September 2026

Topics to be covered

01 Last class in 10 minutes — trigonometry, max–min, phase and superposition formulae

02 Arithmetic progression — common difference, nth term and sum

03 Geometric progression — common ratio, nth term and infinite sum

04 Coordinates, distance and slope

05 Equation of straight line y = mx + c; slope and intercept

06 Draw lines and use them in velocity, ideal-gas and temperature graphs

Trigonometry and max–min — formulae only

180° = π rad **arc = Rθ** **θ must be radian**

sin(A+B) = sinA cosB + cosA sinB

sinθ ≈ θ, tanθ ≈ θ, cosθ ≈ 1 **for θ≤5°, θ in radian**

a sinθ + b cosθ has maximum $\sqrt{(a^2+b^2)}$

$180^\circ = \pi$ rad

$360^\circ = 2\pi$

$60^\circ = ?\ \frac{\pi}{3}$

$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$\tan\left(\frac{\pi}{3}\right) = ?$

What we will learn today

1. Arithmetic progression: identify d, find a term and calculate a sum.

2. Geometric progression: identify r, find a term and calculate an infinite sum.

3. Coordinates, distance and slope.

4. Equation and graph of a straight line.

Rough work / extra work

Series of no

1 2 3 4 5

↳

1 3 5 7 9 → series

↳ Any pattern

1 → 7 → 13 19 25 → Series Pattern

+6 +6 +6 +6

A.P. — equal difference every time

VIP

Arithmetic progression is a sequence in which the difference between consecutive terms is always constant.

1 2nd 3rd nth

a, a+d, a+2d, a+3d, ...

$T_n = a + (n-1)d$	$S_n = n/2\,[2a + (n-1)d]$

a = first term. d = next term − previous term. n = position.

$d = 6$

$a = 1$

$2^{nd} = a + d$

$3^{rd} = a + 2d$

$4^{th} = a + 3d$

$100^{th} = a + 99d \Rightarrow nth = a + (n-1)d$

a_1, a_2, a_3, a_4, a_5

A.P. example: 2, 5, 8, 11, 14, ... $\Rightarrow$ AP

1 2 3

Find a, d, the 10th term and the sum of the first 10 terms.

10th $\Rightarrow n = 10$

$a = ? = 2$ (1st term)

$d = ? = a_2 - a_1 = a_3 - a_2 \Rightarrow$

$= 5 - 2 = 8 - 5$

$d \Rightarrow 3$

$a_n = a + (n-1)d$

$= 2 + (n-1)3 \Rightarrow$

If $n = 10$

$a_{10} = 2 + (10-1) \times 3$

$= 29$

Rough work / extra work

2, 5, 8, 11, 14, 17, 20, 23, 26, 29 | 32

a_{10}

$2 + 5 + 8 + 11 + 14 + 17 + 20 + 23 + 26 + 29$

7 15 26 40 57 77 100 126 155

$S_{10} = 155$

$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

$S_{10} = 5 \times \frac{10}{2}\left[2 \times 2 + (10-1)3\right]$

$= 5\left[4 + 27\right] \rightarrow 5 \times 31 \Rightarrow \boxed{155}$

Simple solution

Step 1: $a=2$ $d=5-2=3$

Step 2: $T_{10} = 2 + 9(3) = 29$

Step 3: $S_{10} = 10/2\ (2+29) = 5\times31 = 155$

$T_{10}=29$ $S_{10}=155$

Question: 7, 11, 15, 19, ...

Find the 10th term and sum of the first 10 terms.

$a_{10} = 43$

$S_{10} = 250$

good

$a = 7, d = 4$

$a_{10} = 7 + 9 \times 4 = 43$

$S_{10} = \frac{n}{2}[2 \times 7 + 9 \times 4]$

$= \frac{5^{1}10}{2}[14 + 36]$

$5 \times [50]$

250

Simple solution

Step 1: a=7 d=4

Step 2: $T_{10} = 7 + 9(4) = 43$

Step 3: $S_{10} = 10/2\ (7+43) = 250$

10th term=43 sum=250

Which term of 27, 24, 21, ... is zero?

1000, 997, 994

Which term of the A.P. 27, 24, 21, ... is zero?

$a_n = 0$

$n = 10$

$a + (n-1)d = 0$

$27 + (n-1)(-3) = 0$

$27 + (-3n) + 3 = 0$

$30 - 3n = 0 \Rightarrow 3n = 30$

$n = 10$

Simple solution

Step 1: $a=27 \qquad d=-3 \qquad T_n=0$

Step 2: $0 = 27 + (n-1)(-3) = 30-3n$

Step 3: $3n=30 \qquad n=10$

The 10th term is zero.

$a_1\ a_2\ a_3\ a_4\ a_5$

Question: 28, 22, x, y, 4

$a + (n-1)d$

28, 22, x, y, 4 is an A.P. Find x and y.

$$a_4 = a + (4-1)d$$
$$= 28 + 3 \times (-6)$$
$$= 28 - 18 = 10$$

Rough work / extra work

$220, 213, 206, 199, 192, \ldots, a_n, 0$

(7) (7)

Find $n = ?$

$a_n = 10$

$a + (n-1)d = 10$

$220 + (n-1)(-7) = 10$

$220 + (-7n + 7) = 10$

$227 - 10 = 7n$

$7n = 217$

$n = \frac{217}{7} = 31$

a_{30} (30) (31)

$a_{30} = 220 + (29)\times(-7)$

$= 220 + (-203)$

$= 17$

$10 - n = d = -7$

$n = -7 - 10 \Rightarrow -17$

Simple solution

Step 1: $a=28 \qquad d=22-28=-6$

Step 2: $x=T_3=28+2(-6)=16$

Step 3: $y=T_4=28+3(-6)=10$

$x=16 \qquad y=10$

Sum of first n natural numbers

Find 1+2+3+...+n.

1. $n^2/2$
2. $n(n+1)/2$
3. $n(n-1)/2$
4. $n^2(n+1)/6$

Sum of first 50 natural no.

1275

$S_{50} = \frac{n}{2}[2a + (n-1)d]$

$S_{50} = 25[2 + 49 \times 1]$

$= 25 \times [51]$

$= 25 \times 51$

For natural no. → a = 1, d = 1

Always

$S_n = \frac{n}{2}[2 + (n-1)] = \frac{n}{2}[1+n]$

51
25
255
102
1275

Simple solution

Step 1: a=1, d=1, last term=n $S_n=n/2(1+n)$

Option 2 n(n+1)/2

Physics question: 25 tuning forks

25 frequencies form an increasing A.P. Difference=10 Hz; 25th frequency=400 Hz. Find the first.

A

$d = 10$

$a_{25} = 400$

$a_1 = ?$

$a_n = a + (n-1)d$

$a_{25} = 400$

$a + 24d = 400$

$a + 240 = 400$

$a = 400 - 240$

$a = 160$

Simple solution

Step 1: $T_{25} = a + 24d$

Step 2: $400 = a + 24(10)$ $\quad a = 400 - 240 = 160$

First tuning-fork frequency = 160 Hz

G.P. — equal ratio every time

$a + d$ = AP

$a \cdot r$ → GP

Geometric progression is a sequence in which the ratio of consecutive terms is constant.

a, ar, ar², ar³, ... $a r^{(n-1)}$

$T_n = ar^{n-1}$	$S_\infty = a/(1-r)$, only if $\lvert r \rvert < 1$

r = any term ÷ previous term.

$a = 2$,

$r = 3$

GP ⇒ $2, 2\times3, 2\times3^2, 2\times3^3, 2\times3^4$

Which sequences are G.P.?

Check the ratio of consecutive terms. Which sequences are G.P.?

A. 2, 4, 8, 12, 24, 48

B. 2, 4, 8, 16, 32, 64

C. 1, 1/2, 1/4, 1/8, ...

D. 1, −1/2, 1/4, −1/8, ...

Which G.P

$r = \frac{1}{2} \rightarrow a \times \frac{1}{2} \Rightarrow \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$\frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

$\frac{-1}{2} \times \frac{-1}{2}$

$r = \frac{-1}{2} \Rightarrow$

$a = 1 \Rightarrow$

$1, \quad 1 \times \frac{-1}{2} \Rightarrow \frac{-1}{2}, \quad \frac{-1}{2} \times \frac{-1}{2} = \frac{1}{4}$

$\frac{1}{4} \times \frac{-1}{2} = -\frac{1}{8}$

Simple solution

sequence	ratio	answer
2,4,8,12,24,48	ratio changes	not G.P.
2,4,8,16,32,64	2	G.P.
1,1/2,1/4,1/8,...	1/2	G.P., S∞=2
1,−1/2,1/4,−1/8,...	−1/2	G.P., S∞=2/3

A sequence is G.P. only if every consecutive ratio is the same.

Rough work / extra work

$a + ar + ar^2 + ar^3$

H.W

Sum $= 2 + 6 + 18 + 54 \Rightarrow 80$

$r = ? = \frac{6}{2} = \frac{18}{6} = 3$

$a = ? = 2$

This a G.P.

$n = 4 \Rightarrow 80 \div 2 = 40$

$n = 3 \Rightarrow 36 \div 2 = 18$

$n = 2 \Rightarrow 8 \div 2 = 4$

$n = 1 \Rightarrow 2 \div 2 = 1$

Find Pattern Here